

Mass Density Profiles in Globular and Open Clusters

A Comparative Study of M2 and M34

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Scientific Motivation

Goal: Measure and compare mass density profiles of two contrasting stellar systems

M2 (NGC 7089)

- Ancient globular cluster
- Age: ~ 13 Gyr
- Distance: 11.5 kpc
- $> 150,000$ stars
- Highly concentrated core

$\text{Fe}/\text{H} \approx -1.6$

M34 (NGC 1039)

- Young open cluster
- Age: ~ 200 Myr
- Distance: 470 pc
- 100–400 members
- Loosely bound
- Solar metallicity

Why study density profiles?

- Encode formation history and dynamical evolution
- Test theoretical models (Plummer, King, Wilson)
- Understand cluster stability and dispersal timescales

Observational Strategy & Signal-to-Noise Requirements

Telescope: Las Cumbres Observatory 0.4m network

Filters: SDSS g' and r' (multi-band for CMD membership)

S/N Calculation:

$$\frac{S}{N} = \frac{FA_{\epsilon}\tau}{\sqrt{N_R^2 + \tau(FA_{\epsilon} + i_{DC} + F_{\beta}A_{\epsilon}\Omega)}}$$

Cluster	Filter	Mag	t_{exp} (s)	S/N	S/N stacked
M2	g'	14	8	180–300	310–520
M2	g'	18	120	45–60	78–104
M2	g'	21.5	300	20–24	35–42
M34	g'	11	10	200–300	350–520
M34	g'	17.5	60	60–80	104–139
M34	g'	19	180	25–30	43–52

Image Stacking: $(S/N)_{\text{stacked}} = \sqrt{N_{\text{exp}}} \times (S/N)_{\text{single}}$

$\Rightarrow$ 3 exposures per field to achieve $S/N > 20$ at faint limit

Methodology: From Images to Density Profiles

1 Photometric Completeness Corrections

- Artificial star tests: inject synthetic stars, measure recovery fraction
- Maximum likelihood fit: $C(m) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{m_{50} - m}{\sqrt{2}\sigma_{\text{comp}}} \right) \right]$
- Richardson-Lucy deconvolution to correct for photometric scatter

2 Multi-dimensional Membership Determination

- **CMD filtering**: Distance to theoretical isochrone
- **Proper motion filtering**: Gaia DR3 coherent cluster motion
- **Spatial filtering**: Centrally concentrated vs. uniform field
- Combined Bayesian probability: $P_{\text{member}} = P_{\text{CMD}} \times P_{\text{PM}} \times P_{\text{spatial}}$

3 Radial Profile Construction

- Bin stars by angular distance from cluster center
- Weight by membership probability
- Fit Plummer model: $\rho(r) = \frac{3M}{4\pi a^3} \left(1 + \frac{r^2}{a^2} \right)^{-5/2}$

Plummer Model Characterization

$$\rho(r) = \frac{3M}{4\pi a^3} \left(1 + \frac{r^2}{a^2}\right)^{-5/2}$$

where a = Plummer radius (core scale), M = total cluster mass

M2 Expectations:

- Small core radius ($a \sim 0.5$ pc)
- High central density
- Steep density profile ($\rho \propto r^{-5}$ at large r)
- Total mass: $M \sim 10^5 M_\odot$
- Dynamically evolved
- Well-fit by Plummer/King models

M34 Expectations:

- Larger core radius ($a \sim 2$ pc)
- Low central density
- Shallow profile
- Total mass: $M \sim 10^3 M_\odot$
- Dynamically young
- Potential tidal disruption signatures

Summary

Project Goals:

- 1 Develop rigorous photometric analysis pipeline for stellar clusters
- 2 Apply multi-dimensional membership determination (CMD + PM + spatial)
- 3 Measure mass density profiles for M2 (globular) and M34 (open cluster)
- 4 Fit Plummer models and characterize structural parameters

Quantitative Requirements:

- **S/N:** >20 at faint limit ($g' \sim 21-22$), >100 for bright members
- **Integration times:** 8–300s per exposure (depends on magnitude)
- **Filters:** SDSS g' and r' for color-magnitude analysis
- **Stacking:** 3 exposures per field $\Rightarrow \sqrt{3} \approx 1.7 \times$ S/N boost
- **Completeness:** $>90\%$ at $m < m_{50}$, measure via artificial star tests

Scientific Impact:

- Understand cluster formation and dynamical evolution

Thank You